

Polynomial Regression Assignment

Energy Consumption Prediction

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Level	Year 3

1. Overview

This assignment covers predicting building energy consumption using polynomial regression built from scratch using gradient descent, alongside Ridge and Lasso regularization via sklearn. The dataset has four input features: Square Footage, Number of Occupants, Appliances Used, and Average Temperature. Three features were selected based on correlation analysis. Models of degree 1 through 9 were trained and compared using Train MSE, Test MSE, Ridge MSE, and Lasso MSE to study the effect of model complexity and regularization.

2. Setup

Parameter	Value
Features Used	Square Footage, Number of Occupants, Appliances Used
Feature Dropped	Average Temperature (correlation = -0.034)
Normalization	StandardScaler (zero mean, unit variance)
Weight Init	sklearn defaults
Ridge Alpha	150
Lasso Alpha	150
Error Metric	MSE
Train/Test Split	80% / 20% (random_state=42)
Degree Range	1 to 9

3. MSE Results

Train MSE, Test MSE, Ridge MSE, and Lasso MSE for each polynomial degree:

Degree	Train MSE	Test MSE	Ridge MSE	Lasso MSE
1	166,767.52	170,679.16	184,784.26	208,916.27
2	165,249.18	172,449.46	185,366.88	208,916.27
3	164,357.36	174,818.80	190,979.87	239,506.22
4	161,888.17	178,200.77	194,277.89	239,506.22
5	156,530.99	182,432.96	188,239.17	262,956.71
6	152,839.98	195,267.59	190,540.92	263,633.19
7	145,864.20	205,489.71	190,886.05	270,702.89
8	136,864.88	270,268.35	201,844.21	271,806.78
9	125,873.98	426,771.29	212,811.94	277,997.60

4. Plots

Degree vs MSE (Train, Test, Ridge, Lasso)

Train MSE decreases steadily with degree, dropping from ~166k at degree 1 to ~125k at degree 9. This is expected — more complex models fit the training data better. Test MSE tells the opposite story: it starts at ~170k and rises sharply, reaching ~426k at degree 9, which is clear overfitting. Ridge regularization keeps test error controlled, peaking at ~212k at degree 9 compared to ~426k for unregularized. Lasso is the most aggressive, pushing MSE higher from the start due to its tendency to zero out features entirely.

5. Conclusion

Best degree: Degree 1 achieves the lowest test MSE of ~170,679. The dataset has predominantly linear relationships as confirmed by the correlation heatmap, so additional polynomial terms only introduce overfitting without capturing real signal.

Overfitting: The gap between train and test MSE widens dramatically at higher degrees. At degree 9, train MSE is ~125k while test MSE explodes to ~426k — a clear sign the model has memorized the training data and fails to generalize.

Ridge regularization: Ridge significantly controls overfitting at high degrees. At degree 9 it achieves ~212k compared to ~426k for unregularized, showing that penalizing large coefficients helps the model generalize better. At degree 1, Ridge performs slightly worse (~184k vs ~170k) since regularization is unnecessary when the model is already simple.

Lasso regularization: Lasso performs worse than Ridge across all degrees in this case. With $\alpha=150$, Lasso aggressively zeroes out coefficients and underfits even at degree 1 (~208k). This suggests the features all carry useful signal and should not be eliminated, making Ridge a better choice for this dataset.

Feature selection: Average Temperature was dropped due to its near-zero correlation of -0.034 with the target. The three retained features are mutually uncorrelated, and Square Footage dominates with a correlation of 0.77, explaining most of the predictive power of the model.